

Matthew J. Nitzken

Principal Machine Learning Engineer

Remote - Louisville, KY

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About

I'm a machine learning engineer and data scientist with a background in electrical, computer, and bioengineering, and more than 15 years working in AI. I'm currently a Principal Machine Learning Engineer at DataRobot, where I lead the engineering behind our LLM agents and agentic workflows and build generative AI products. It's been the near-total focus of my recent work, and the direction I want to keep building in. The rest of my career has been full-stack machine learning and data science in large-scale data environments, not just building models but turning new ideas into real, shipped products, and I gravitate toward hard problems that haven't been solved yet and getting them into production. Along the way I've worked across generative AI, LLMs, and agents; time series and deep learning; healthcare; data engineering; medical diagnostics and research; image and signal processing; technical publishing; software development; and controls automation. I enjoy collaborating with, learning from, and mentoring the people around me.

Skills

Python	Machine Learning	Deep Learning	Time Series
LLMs & Agents	Generative AI	Model Explainability	Big Data
Data Engineering	Cloud & Kubernetes	Test-Driven Development	Mentorship & Leadership

Education

University of Louisville

Doctor of Philosophy (Ph.D.), in Electrical and Computer Engineering

[Louisville, Kentucky](#)

2010 – 2015

- Dissertation: Shape Analysis of the Human Brain

University of Louisville

Masters of Engineering (M.Eng.), in Bioengineering

[Louisville, Kentucky](#)

2009 – 2010

University of Louisville

Bachelor of Science (B.S.), in Bioengineering

[Louisville, Kentucky](#)

2005 – 2009

Experience

DataRobot

[Louisville, KY — Remote](#)

Data Scientist → Senior → Staff → Principal Machine Learning Engineer

Jul 2019 – Present

Engineering lead for DataRobot's LLM agent and agentic workflow products, after growing from data scientist across time series, model explainability, platform architecture, and generative AI.

- Lead the engineering for DataRobot's LLM agents and agentic workflows, and build the generative AI and multi-agent products — including development templates that let teams run frameworks like LangGraph, CrewAI, and LlamaIndex inside DataRobot.
- Co-architected af-components, a framework for building agentic generative AI solutions: drop-in templates carry their own Terraform/Pulumi infrastructure so a full solution deploys in one command, and central fixes for bugs, CVEs, and features push out to internal and external users automatically.
- Joined the in-flight Universal SHAP effort and carried it to delivery: because SHAP doesn't natively work on time series data, developed and validated a new method to make it work, then led the engineering to ship it — putting SHAP explanations on 200+ model blueprints that had none, across DataRobot SaaS (single- and multi-tenant) and on-prem.
- Reworked a stalled plan to break up a core platform monolith into one that could actually be implemented — restructuring test infrastructure, splitting code into new repositories with their own CI/CD, and reworking core code; the resulting approach was picked up and reused across other teams.

- Created and led several core time series products — incremental (online) learning that removed training-size limits, Segmented Modeling for large intermixed datasets, and an unsupervised clustering framework spanning feature processing, modeling, insights, and predictions.
- Built DataRobot's early time series trust and explainability features — feature importance, feature impact, and drift — adapting AutoML methods into then-unexplored territory, on top of large-scale data-warehousing and feature-engineering infrastructure I designed.
- *Experience focused on Python, LLMs, LangGraph, TensorFlow, Keras, Docker, Kubernetes, Helm, AWS, GCP, Snowflake, Postgres, MongoDB, and TDD*

Humana

Louisville, KY / Boston, MA

Analytics Consultant → Lead Data Scientist

Oct 2016 – Jul 2019

Built and led machine learning across Humana's clinical and Medicare STARS analytics, moving from analytics consultant to lead data scientist across three teams.

- Replaced Humana's Medicare STARS rating forecasts — previously unreliable ARIMA projections — with Monte Carlo simulations (one per plan and measure, plus a global model) that produced a probability curve across the 1–5 STARS scale, showing leadership where each plan was likely to land, its odds of slipping, and where to focus money and resources.
- Built a large set of models predicting member health trajectories across conditions common to Medicare members (diabetes, COPD, flu vaccination), ranging from straightforward ML to SAS/Spark-fronted pipelines to TensorFlow deep learning.
- Later joined a new Boston team building real-time deep learning for Medicare members — migrating data off on-prem to GCP and building the data pipelines and front-end-backed applications around the models.
- *Experience using Python, TensorFlow, Keras, SkLearn, SAS, GCP, AWS, Hadoop, SQL, Airflow, and TDD*

Alliant Technologies

Louisville, KY

Software Engineer → Lead Software and Controls Engineer

May 2015 – Oct 2016

Software and controls developer for automated high-speed sorter systems, promoted to lead the software development and commissioning teams.

- Led software development and controls commissioning through build-out and QA on large-scale material-handling projects.
- Built an Ethernet/IP communications stack for high-speed sortation, and a Python server that simulated the control server so new systems could be tested before deployment.

NeuroAtlas

Louisville, KY

Founder and Machine Learning Engineer

Mar 2013 – May 2015

Co-founder and lead machine learning engineer at NeuroAtlas, building neurological diagnostic software for autism spectrum disorders.

- Went through two healthcare accelerators (XLerateHealth, LaunchIT) and ran several grant-funded initiatives; co-led direction, design, and fundraising.

University of Louisville

Louisville, KY

Graduate Researcher — Bioluminescence Laboratory

Jul 2010 – May 2015

Ph.D. research developing machine learning for digital diagnostics.

- Developed and published 2D/3D shape, image-, and signal-processing algorithms and computer-aided diagnostic ML, across autism, dyslexia, cancer, Alzheimer's, and cardiac/EMG/ECG signals.
- Presented research at national and international conferences, through talks and posters.

Publications and Patents

Please see publications.mattnitzken.com for an up-to-date list of my publications and patents.